

Kaiwen Wu

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Research

I am interested in both theory and applications of machine learning. I have been working on various areas in probabilistic machine learning including Gaussian processes, variational inference, Bayesian optimization, and active learning. Before starting my PhD, I used to work on generative modeling, convex optimization, and adversarial robustness of deep learning models.

Education

University of Pennsylvania Doctor of Philosophy in Computer and Information Science • Advisor: Jacob R. Gardner	Philadelphia, PA 2021 - Present
University of Waterloo Master of Mathematics in Computer Science (Thesis Option) • Advisor: Yaoliang Yu	Waterloo, ON 2018 - 2020
Nanjing University Bachelor of Science in Computer Science	Nanjing, China 2014 - 2018

Work Experience

Meta Reality Labs Research Scientist Intern • Worked on experimental designs for VR/AR hardware using active learning.	Redmond, WA Aug 2024 - Dec 2024
Borealis AI Research Intern • Worked on robust mean estimation on noisy data using generative adversarial networks. • Published a paper at AISTATS 2020.	Waterloo, ON Jan 2019 - May 2019

Open Source Contributions

BoTorch (<https://github.com/pytorch/botorch>)
A Python package for Bayesian optimization in PyTorch
• Implemented an improved elliptical slice sampling method for truncated normal distributions.

Publications

Refereed Conference Proceedings

- [1] **Kaiwen Wu** and Jacob R. Gardner
Understanding Stochastic Natural Gradient Variational Inference
Proceedings of the 41st International Conference on Machine Learning (ICML). 2024.
- [2] **Kaiwen Wu**, Jonathan Wenger, Haydn T Jones, Geoff Pleiss, and Jacob Gardner
Large-Scale Gaussian Processes via Alternating Projection
Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS). 2024.
- [3] Kyurae Kim, Jisu Oh, **Kaiwen Wu**, Yian Ma, and Jacob R. Gardner
On the Convergence of Black-Box Variational Inference
Advances in Neural Information Processing Systems (NeurIPS). 2023.
- [4] Natalie Maus, **Kaiwen Wu**, David Eriksson, and Jacob Gardner
Discovering Many Diverse Solutions with Bayesian Optimization
Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS). 2023.

- [5] **Kaiwen Wu**, Kyurae Kim, Roman Garnett, and Jacob R. Gardner
The Behavior and Convergence of Local Bayesian Optimization
Advances in Neural Information Processing Systems (NeurIPS). 2023.
- [6] Quan Nguyen, **Kaiwen Wu**, Jacob Gardner, and Roman Garnett
Local Bayesian optimization via maximizing probability of descent
Advances in Neural Information Processing Systems (NeurIPS). 2022.
- [7] **Kaiwen Wu**, Gavin Weiguang Ding, Ruitong Huang, and Yaoliang Yu
On Minimax Optimality of GANs for Robust Mean Estimation
Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS). 2020.
- [8] **Kaiwen Wu**, Allen Wang, and Yaoliang Yu
Stronger and Faster Wasserstein Adversarial Attacks
Proceedings of the 37th International Conference on Machine Learning (ICML). 2020.

Lightly Reviewed Papers

- [9] **Kaiwen Wu** and Jacob R. Gardner
A Fast, Robust Elliptical Slice Sampling Method for Linearly Truncated Multivariate Normal Distributions
NeurIPS Workshop on Bayesian Decision-making and Uncertainty. 2024.
- [10] Guojun Zhang, **Kaiwen Wu**, Pascal Poupart, and Yaoliang Yu
Newton-type Methods for Minimax Optimization
ICML Workshop on Beyond First-order Methods in Machine Learning Systems. 2021.

Teaching Experience

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| • Teaching Assistant, University of Pennsylvania, CIS 520 Machine Learning | FA 2023 |
| • Teaching Assistant, University of Pennsylvania, ESE 204 Decision Models | FA 2022 |
| • Teaching Assistant, University of Waterloo, CS 370 Numerical Computation | SP 2020 |
| • Teaching Assistant, University of Waterloo, CS 486/686 Introduction to Artificial Intelligence | SP 2019 |
| • Teaching Assistant, University of Waterloo, CS 480/680 Introduction to Machine Learning | FA 2018 |

Honors & Awards

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| • NeurIPS Top Reviewer | 2024 |
| • Vector Scholarship in Artificial Intelligence, Vector Institute | 2018 |
| • Entrance Scholarship, University of Waterloo | 2018 |
| • Jimin Liu Scholarship, Nanjing University | 2017 |

Academic Service

Conference Reviews

- Program Committee Member, AAAI Conference on Artificial Intelligence (2021)
- Reviewer, Conference on Neural Information Processing Systems (2023 – 2024)
- Reviewer, International Conference on Artificial Intelligence and Statistics (2021, 2024)
- Reviewer, International Conference on Learning Representations (2024 – 2025)
- Reviewer, International Conference on Machine Learning (2023 – 2025)

Journal Reviews

- Reviewer, Transactions on Machine Learning Research (2025)